Vector-Valued Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Position, velocity, and acceleration vectors**

A particle's position is given by $\vec{r}(t) = \langle t^2, t^3 \rangle$. Find the velocity vector $\vec{v}(t)$ and acceleration vector $\vec{a}(t)$.

$\vec{v}(t) = \langle 2t, 3t^2 \rangle$. $\vec{a}(t) = \langle 2, 6t \rangle$.

2 **[Foundational]** • **Target: Position, velocity, and acceleration vectors**

For the same particle, find the speed at $t = 2$.

$\vec{v}(2) = \langle 4, 12 \rangle$. Speed $= |\vec{v}(2)| = \sqrt{4^2 + 12^2} = \sqrt{16 + 144} = \sqrt{160} = 4\sqrt{10}$.

3 **[Foundational]** • **Target: Integrating vector-valued functions**

Given $\vec{v}(t) = \langle 3t^2, 2t \rangle$ and initial position $\vec{r}(0) = \langle 1, 4 \rangle$, find $\vec{r}(t)$.

$\vec{r}(t) = \langle t^3 + C_1, t^2 + C_2 \rangle$. Using $\vec{r}(0) = \langle 1, 4 \rangle$: $C_1 = 1$, $C_2 = 4$. $\vec{r}(t) = \langle t^3 + 1, t^2 + 4 \rangle$.

4 **[Foundational]** • **Target: Total distance traveled**

A particle has velocity $\vec{v}(t) = \langle \cos t, \sin t \rangle$ for $t \in [0, \pi]$. Find the total distance traveled.

Speed $= |\vec{v}(t)| = \sqrt{\cos^2 t + \sin^2 t} = 1$ (constant). Distance $= \int_0^{\pi} 1 \, dt = \pi$.

5 **[Foundational]** • **Target: Vectors and parametric motion combined**

A particle moves along $x(t) = e^t$, $y(t) = e^{-t}$. Find the velocity vector and speed at $t = 0$.

$\vec{v}(t) = \langle e^t, -e^{-t} \rangle$. At $t = 0$: $\vec{v}(0) = \langle 1, -1 \rangle$. Speed $= \sqrt{1^2 + (-1)^2} = \sqrt{2}$.

6 **[Foundational]** • **Target: Vectors and parametric motion combined**

For the particle above, find the magnitude of the acceleration vector at $t = 0$.

$\vec{a}(t) = \langle e^t, e^{-t} \rangle$. At $t = 0$: $\vec{a}(0) = \langle 1, 1 \rangle$. $|\vec{a}(0)| = \sqrt{1^2 + 1^2} = \sqrt{2}$.

7 **[Foundational]** • **Target: Unit tangent vectors**

For $\vec{r}(t) = \langle t, t^2 \rangle$, find the unit tangent vector $\vec{T}(t)$ at $t = 1$.

$\vec{v}(t) = \langle 1, 2t \rangle$; at $t = 1$: $\vec{v}(1) = \langle 1, 2 \rangle$, $|\vec{v}(1)| = \sqrt{1 + 4} = \sqrt{5}$. $\vec{T}(1) = \left\langle \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Unit tangent vectors

Explain why $\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$ always has magnitude 1.

Dividing any nonzero vector by its own magnitude produces a vector with magnitude $\frac{|\vec{v}(t)|}{|\vec{v}(t)|} = 1$ by definition — this is the standard construction of a unit vector in the same direction.

9 [Intermediate] • Target: Motion on a circle and centripetal acceleration

A particle moves along $\vec{r}(t) = \langle 3\cos t, 3\sin t \rangle$. Find $\vec{v}(t)$, $\vec{a}(t)$, and verify that $\vec{a}(t)$ points toward the origin (is a scalar multiple of $-\vec{r}(t)$).

$\vec{v}(t) = \langle -3\sin t, 3\cos t \rangle$. $\vec{a}(t) = \langle -3\cos t, -3\sin t \rangle = -\vec{r}(t)$. Since $\vec{a}(t)$ is exactly -1 times $\vec{r}(t)$, it points directly toward the origin (centripetal acceleration).

10 [Intermediate] • Target: Motion on a circle and centripetal acceleration

For the same particle, verify that $\vec{v}(t) \cdot \vec{r}(t) = 0$ for all t, and interpret this geometrically.

$\vec{v}(t) \cdot \vec{r}(t) = (-3\sin t)(3\cos t) + (3\cos t)(3\sin t) = -9\sin t\cos t + 9\sin t\cos t = 0$. This confirms the velocity vector is always perpendicular to the position vector — consistent with circular motion, where velocity is tangent to the circle.

11 [Intermediate] • Target: Displacement vs. distance for vector motion

A particle has position $\vec{r}(t) = \langle t^2, t \rangle$ for $t \in [0, 2]$. Find the displacement vector from $t = 0$ to $t = 2$.

$\vec{r}(0) = \langle 0, 0 \rangle$; $\vec{r}(2) = \langle 4, 2 \rangle$. Displacement $= \vec{r}(2) - \vec{r}(0) = \langle 4, 2 \rangle$.

12 [Intermediate] • Target: Displacement vs. distance for vector motion

Set up (but do not evaluate) the integral for the total distance traveled by the particle $\vec{r}(t) = \langle t^2, t \rangle$ over $t \in [0, 2]$.

$\vec{v}(t) = \langle 2t, 1 \rangle$. Distance $= \int_0^2 |\vec{v}(t)|\, dt = \int_0^2 \sqrt{(2t)^2 + 1^2}\, dt = \int_0^2 \sqrt{4t^2 + 1}\, dt$.

13 [Intermediate] • Target: Vector operations review

Find the dot product $\vec{u} \cdot \vec{w}$ for $\vec{u} = \langle 3, -2 \rangle$ and $\vec{w} = \langle -1, 4 \rangle$, and state whether the vectors are perpendicular.

$\vec{u} \cdot \vec{w} = 3(-1) + (-2)(4) = -3 - 8 = -11$. Since the dot product is not 0, the vectors are NOT perpendicular.

14 [Intermediate] • Target: Curvature-related speed problems

A particle moves along $\vec{r}(t) = \langle t^3, t^2 \rangle$. Find the times t when the particle's speed is exactly $\sqrt{13}$.

$\vec{v}(t) = \langle 3t^2, 2t \rangle$. Speed$^2 = 9t^4 + 4t^2 = 13$. Let $w = t^2$: $9w^2 + 4w - 13 = 0$. Using the quadratic formula: $w = \frac{-4 \pm \sqrt{16 + 468}}{18} = \frac{-4 \pm 22}{18}$. Taking the positive root: $w = 1$, so $t^2 = 1 \Rightarrow t = \pm 1$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For the particle $\vec{r}(t) = \langle t, \ln t \rangle$ $(t > 0)$, find the velocity vector and speed at $t = 1$.

$\vec{v}(t) = \left\langle 1, \frac{1}{t} \right\rangle$. At $t = 1$: $\vec{v}(1) = \langle 1, 1 \rangle$. Speed $= \sqrt{1^2 + 1^2} = \sqrt{2}$.

16 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 6.02 Taylor and Maclaurin Series

A projectile is launched with position $\vec{r}(t) = \langle 20t, 30t - 4.9t^2 \rangle$ (meters, seconds). Find the velocity vector $\vec{v}(t)$ and the time when the projectile reaches maximum height (where the vertical velocity component is 0).

$\vec{v}(t) = \langle 20, 30 - 9.8t \rangle$. Maximum height occurs when $30 - 9.8t = 0 \Rightarrow t = \frac{30}{9.8} \approx 3.06$ s.

17 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 1.01 Integration by Parts

For the same projectile, find the speed at $t = 0$ (launch) and interpret its two components.

$\vec{v}(0) = \langle 20, 30 \rangle$. Speed $= \sqrt{20^2 + 30^2} = \sqrt{400 + 900} = \sqrt{1300} \approx 36.1$ m/s. The horizontal component (20 m/s) stays constant throughout the flight, while the vertical component (30 m/s initially) decreases due to gravity.

18 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

A particle moves along $\vec{r}(t) = \langle t^3 - 3t,\ t^2 - 4 \rangle$. Find all times t when the particle is momentarily at rest (both velocity components equal 0 simultaneously).

$\vec{v}(t) = \langle 3t^2 - 3, 2t \rangle$. Need $3t^2 - 3 = 0$ AND $2t = 0$ simultaneously. From the second equation, $t = 0$; check the first: $3(0)^2 - 3 = -3 \neq 0$. There is NO time when both components are zero at once, so the particle is never fully at rest.

19 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

For the same particle, find the times when only the horizontal velocity component is 0.

$3t^2 - 3 = 0 \Rightarrow t^2 = 1 \Rightarrow t = \pm 1$.

20 [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 3.01 Parametric Equations

State the arc length formula for a vector-valued function $\vec{r}(t) = \langle x(t), y(t) \rangle$ on $[a, b]$, and explain how it relates to the speed $|\vec{v}(t)|$.

$L = \int_a^b |\vec{v}(t)|\, dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$ — arc length is the integral of speed over time, which matches the intuition that distance traveled equals the accumulated speed.

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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Position, velocity, and acceleration vectors

A particle moves along $\vec{r}(t) = \langle 5t^2, 12t^2 \rangle$. Find the magnitude of the (constant) acceleration vector $|\vec{a}(t)|$.

a. 26 ✓ **b.** 28

c. 34 **d.** 13

$\vec{a}(t) = \langle 10, 24 \rangle$ (constant). $|\vec{a}(t)| = \sqrt{10^2 + 24^2} = \sqrt{676} = 26$.

22 [Foundational] • Target: Integrating vector-valued functions

A particle moves along $\vec{r}(t) = \langle t^3 - 12t, t^2 + 3 \rangle$. Find the positive time t when the horizontal velocity component is 0.

a. -2 **b.** 6

c. 2 ✓ **d.** 3

$\vec{v}(t) = \langle 3t^2 - 12, 2t \rangle$. Setting $3t^2 - 12 = 0$: $t^2 = 4 \Rightarrow t = \pm 2$. The positive value is 2.

23 [Foundational] • Target: Total distance traveled

A particle moves along $\vec{r}(t) = \langle 7t, 48t - 4t^2 \rangle$. Find the time t when the vertical velocity component is 0 (the time of maximum height).

a. 7 **b.** 6 ✓

c. 12 **d.** 5

$\vec{v}(t) = \langle 7, 48 - 8t \rangle$. Setting $48 - 8t = 0$: $t = \frac{-(48)}{-8} = 6$.

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24 [Foundational] • Target: Vectors and parametric motion combined

Consider the function $f(x) = e^x \sin(x)$. A particle's position on a number line is given by $s(t) = f(t)$ for $t \geq 0$. The particle is momentarily at rest when its velocity is zero. The acceleration of the particle at these moments is given by $a(t) = f''(t)$. Find the smallest positive value of t at which the particle is momentarily at rest, and then determine the acceleration of the particle at that exact moment.

a. **Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\sqrt{2}e^{\frac{3\pi}{4}}$** ✓

b. Rest at $t = \frac{\pi}{2}$, acceleration $a = e^{\frac{\pi}{2}}$

c. Rest at $t = \frac{3\pi}{4}$, acceleration $a = -\frac{\sqrt{2}}{2}e^{\frac{3\pi}{4}}$

d. Rest at $t = \pi$, acceleration $a = 0$

First, we need to find the velocity, which is the first derivative of the position function $s(t) = f(t) = e^t \sin(t)$. Using the product rule, $v(t) = f'(t) = e^t \sin(t) + e^t \cos(t) = e^t(\sin(t) + \cos(t))$. The particle is momentarily at rest when $v(t) = 0$. Since $e^t > 0$ for all t, we need $\sin(t) + \cos(t) = 0$, which implies $\tan(t) = -1$. The general solutions for this are $t = \frac{3\pi}{4} + n\pi$, where n is an integer. For $t \geq 0$, the smallest positive value occurs when $n = 0$, so.

25 [Foundational] • Target: Unit tangent vectors

A particle moves along the x-axis. Its velocity is given by the function $v(t) = \sqrt{t^2 + 1}$ for $t \geq 0$. The position of the particle at time $t = 0$ is $x(0) = 5$. Consider the function $P(x)$ defined as the total displacement of the particle from $t = 0$ to $t = x$. That is, $P(x) = x(x) - x(0)$. Which of the following statements about $P(x)$ is true for $x > 0$?

a. **$P(x)$ is always positive and increasing.** ✓

b. $P(x)$ is always negative and decreasing.

c. $P(x)$ is always positive and decreasing.

d. $P(x)$ is always negative and increasing.

The velocity of the particle is given by $v(t) = \sqrt{t^2 + 1}$. Since $t \geq 0$, $t^2 \geq 0$, so $t^2 + 1 \geq 1$. Therefore, $v(t) = \sqrt{t^2 + 1} \geq \sqrt{1} = 1$ for all $t \geq 0$. This means the velocity is always positive. The position of the particle is given by $x(t)$. The Fundamental Theorem of Calculus states that the change in position (displacement) is the integral of the velocity function: $x(t) - x(0) = \int_0^t v(u)\,du$. The function $P(x)$ represents the total displacement from $t = 0$ to.

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26 [Foundational] • Target: Motion on a circle and centripetal acceleration

Unit tangent vector $\mathbf{T}(t) = \mathbf{r}'(t)/|\mathbf{r}'(t)|$. It has magnitude:

a. 1 ✓

b. 0

c. $|r'|$

d. t

Applying Vectors and Vector-Valued Functions principles, Unit tangent: $|\mathbf{T}| = 1$. Therefore, the verified correct option is 1.

27 [Foundational] • Target: Displacement vs. distance for vector motion

For a circle of radius R, curvature $\kappa =$

a. R

b. $2R$

c. $1/R^2$

d. $1/R$ ✓

Applying Vectors and Vector-Valued Functions principles, Circle: $\kappa = 1/R$. Therefore, the verified correct option is $1/R$.

28 [Foundational] • Target: Vector operations review

Tangent plane at $(0, 0, 1)$ on unit sphere: $z = 1$. Normal direction:

a. $(0, 1, 0)$

b. $(1, 0, 0)$

c. $(0, 0, 1)$ ✓

d. $(1, 1, 1)$

North pole: outward normal $= (0, 0, 1)$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 **[Intermediate]** • **Target: Curvature-related speed problems**

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ **and** $t = 3$ ✓

The particle's position is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$. The velocity is the derivative of the position with respect to time, $v(t) = z'(t)$. Differentiating the real and imaginary parts separately: Real part of $z'(t)$: $\frac{d}{dt}(2t^2 - 3t + 1) = 4t - 3$ Imaginary part of $z'(t)$: $\frac{d}{dt}(t^3 - 6t^2 + 11t - 6) = 3t^2 - 12t + 11$ So, $v(t) = (4t - 3) + i(3t^2 - 12t + 11)$. The particle is at rest when $v(t) = 0$. This means both the real and imaginary parts of $v(t)$ must be.

30 **[Intermediate]** • **Target: Projectile motion as a vector-valued function**

A particle's velocity is given by $v(t) = 3t^2 - 12t + 9$ for $t \geq 0$. The position of the particle at time $t = 0$ is $s(0) = 5$. \\ Consider a second particle whose position $p(t)$ is described by the integral $p(t) = \int_0^t v(u)\,du$. \\ Find the net displacement of the first particle from $t = 1$ to $t = 3$.

a. -4 ✓

b. 4

c. 8

d. -8

The net displacement of the first particle from $t = 1$ to $t = 3$ is given by the definite integral of its velocity function over this interval: $\Delta s = \int_1^3 v(t)\,dt$. \\ We are given $v(t) = 3t^2 - 12t + 9$. \\ To find the definite integral, we first find the antiderivative of $v(t)$: \\ $s(t) = \int (3t^2 - 12t + 9)\,dt = t^3 - 6t^2 + 9t + C$. \\ Since the initial position is $s(0) = 5$, we have $0^3 - 6(0)^2 + 9(0) + C = 5$, which means $C = 5$. Thus, $s(t) = t^3 - 6t^2 + 9t + 5$. \\ The net displace.

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31 [Intermediate] • Target: Finding when a particle is at rest

The velocity of a particle moving along a straight line is given by the function $v(t) = t^2 - 6t + 8$ m/s. What is the instantaneous acceleration of the particle at time $t = 3$ seconds?

a. -1 m/s^2

b. 0 m/s^2

c. 1 m/s^2

d. -2 **m/s^2** ✓

The velocity of the particle is given by $v(t) = t^2 - 6t + 8$. Acceleration is the rate of change of velocity with respect to time, which means acceleration is the derivative of the velocity function, $a(t) = v'(t)$. Differentiating $v(t)$ with respect to t, we get: $a(t) = \frac{d}{dt}(t^2 - 6t + 8) = 2t - 6$. We need to find the instantaneous acceleration at time $t = 3$ seconds. Substituting $t = 3$ into the acceleration function: $a(3) = 2(3) - 6 = 6 - 6 = 0$ m/s^2. Therefore, the instantaneous acceleratio.

32 [Intermediate] • Target: Vector-valued functions and arc length

A particle's velocity is given by the function $v(t) = 5$ for $0 \le t \le 4$. Find the total displacement of the particle over this time interval.

a. 10

b. 15

c. **20** ✓

d. 25

The displacement of the particle is the definite integral of its velocity function over the given time interval. \\ Displacement = $\int_0^4 v(t)\,dt = \int_0^4 5\,dt$. \\ To evaluate this definite integral, we find the antiderivative of 5, which is $5t$. \\ Then, we evaluate the antiderivative at the limits of integration: \\ $[5t]_0^4 = 5(4) - 5(0) = 20 - 0 = 20$. \\ Therefore, the total displacement of the particle is 20 units.

33 [Intermediate] • Target: Position, velocity, and acceleration vectors

Distance between skew lines $L_1 : \mathbf{r} = \mathbf{a} + s\mathbf{d}_1$ and $L_2 : \mathbf{r} = \mathbf{b} + t\mathbf{d}_2$:

a. $|\mathbf{b} - \mathbf{a}|$

b. $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$ ✓

c. $|\mathbf{d}_1 - \mathbf{d}_2|$

d. 0

Applying Vectors and Vector-Valued Functions principles, Skew line distance formula. Therefore, the verified correct option is $|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{d}_1 \times \mathbf{d}_2)|/|\mathbf{d}_1 \times \mathbf{d}_2|$.

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34 [Intermediate] • Target: Integrating vector-valued functions

Parametric surface $\mathbf{r}(u,v) = (u\cos v, u\sin v, u^2)$. At $(u,v) = (1,0)$:

a. tangent plane is $z = x^2$

b. **tangent plane is $z = 2x - 1$** ✓

c. normal is $(0,0,1)$

d. surface is a sphere

At $(u,v) = (1,0)$, the point is $(1,0,1)$, $\mathbf{r}_u = (1,0,2)$, and $\mathbf{r}_v = (0,1,0)$, giving normal $\mathbf{r}_u \times \mathbf{r}_v = (-2,0,1)$ and tangent plane $-2(x-1) + z - 1 = 0 \Rightarrow z = 2x - 1$.

35 [Intermediate] • Target: Total distance traveled

Find the curvature of $\mathbf{r}(t) = (t, t^2, 0)$ at $t = 0$:

a. 2 ✓

b. 1

c. 0

d. 1/2

$\kappa = \frac{|\mathbf{r}' \times \mathbf{r}''|}{|\mathbf{r}'|^3}$; $\mathbf{r}' = (1, 2t, 0)$, $\mathbf{r}'' = (0,2,0)$; at $t = 0$: $|(1,0,0) \times (0,2,0)|/|(1,0,0)|^3 = |(0,0,2)|/1 = 2$.

36 [Intermediate] • Target: Vectors and parametric motion combined

For $\mathbf{r}(t) = \langle 2t, 3t^2, t\rangle$, find the normal component of acceleration at $t = 1$.

a. $\frac{6\sqrt{205}}{41}$ ✓

b. $\frac{36\sqrt{41}}{41}$

c. 6

d. 36

$\mathbf{v} = \langle 2,6,1\rangle$, $\mathbf{a} = \langle 0,6,0\rangle$. $a_N = \frac{||\mathbf{v} \times \mathbf{a}||}{||\mathbf{v}||} = \frac{6\sqrt{205}}{41}$.

37 [Intermediate] • Target: Unit tangent vectors

For $\mathbf{r}(t) = \langle 2t, t^2, t\rangle$, find the tangential component of acceleration at $t = 2$.

a. $\frac{2\sqrt{105}}{21}$

b. $\frac{8\sqrt{21}}{21}$ ✓

c. 2

d. 8

$\mathbf{v} = \langle 2,4,1\rangle$, $\mathbf{a} = \langle 0,2,0\rangle$. $a_T = \frac{\mathbf{v} \cdot \mathbf{a}}{||\mathbf{v}||} = \frac{8\sqrt{21}}{21}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Motion on a circle and centripetal acceleration • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

The torsion τ of a space curve measures how fast the curve twists out of its osculating plane. For a helix $\mathbf{r} = (a\cos t, a\sin t, bt)$, as $b \to 0$ (curve flattens to a circle), $\tau \to$

a. $1/a$

b. 0 ✓

c. ∞

d. b

Applying Vectors and Vector-Valued Functions principles, $\tau = b/(a^2 + b^2) \to 0$ as $b \to 0$. Therefore, the verified correct option is 0.

39 [Challenging] • Target: Displacement vs. distance for vector motion • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Distance from point $P = (1, 2, 3)$ to plane $2x + y - 2z = 5$:

a. $\frac{7}{3}$ ✓

b. $\frac{2}{3}$

c. 1

d. 3

Using the point-to-plane distance formula, $d = \frac{|2(1) + 1(2) - 2(3) - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{|-7|}{3} = \frac{7}{3}$.

40 [Challenging] • Target: Vector operations review • Cross-Lesson: 3.01 Parametric Equations

Divergence theorem: $\oiint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (abla \cdot \mathbf{F})\, dV$. For $\mathbf{F} = \mathbf{r} = (x, y, z)$, $abla \cdot \mathbf{F} = 3$: flux through unit sphere =

a. 3

b. $3V = 4\pi$ ✓

c. 4π

d. 12π

Applying Vectors and Vector-Valued Functions principles, $3 \cdot V = 3 \cdot 4\pi/3 = 4\pi$. Therefore, the verified correct option is $3V = 4\pi$.

41 [Challenging] • Target: Curvature-related speed problems • Cross-Lesson: 4.01 Polar Curves and Derivatives

Position vector of midpoint of segment from $A(1, 2, 3)$ to $B(5, 6, 7)$:

a. $(6, 8, 10)$

b. $(2, 2, 2)$

c. $(3, 4, 5)$ ✓

d. $(4, 4, 4)$

Applying Vectors and Vector-Valued Functions principles, Midpoint $= (A + B)/2 = (3, 4, 5)$. Therefore, the verified correct option is $(3, 4, 5)$.

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42 [Challenging] • Target: Projectile motion as a vector-valued function • Cross-Lesson: 4.02 Area in Polar Coordinates

Curvature of helix $\mathbf{r} = (a\cos t, a\sin t, bt)$:

a. ab

b. $b/(a^2+b^2)$

c. $1/a$

d. $a/(a^2+b^2)$ ✓

Applying Vectors and Vector-Valued Functions principles, $\kappa = a/(a^2+b^2)$. Therefore, the verified correct option is $a/(a^2+b^2)$.

43 [Challenging] • Target: Finding when a particle is at rest • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

Potential function for $\mathbf{F} = (2xy, x^2+3y^2, 0)$:

a. $f = x^2y + y^3$ ✓

b. $f = x^2y^2$

c. $f = xy^2$

d. $f = x^2 + y^2$

$f_x = 2xy \Rightarrow f = x^2y + g(y)$; $f_y = x^2 + g'(y) = x^2 + 3y^2 \Rightarrow g' = 3y^2 \Rightarrow g = y^3$.

44 [Challenging] • Target: Vector-valued functions and arc length • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Find the distance from $(2,\ -2,\ 0)$ to the plane $-3y + z = -4$.

a. 10

b. $\frac{\sqrt{10}}{5}$

c. 1

d. $\sqrt{10}$ ✓

$D = \frac{|ax_0 + by_0 + cz_0 - d|}{\sqrt{a^2+b^2+c^2}} = \frac{10}{\sqrt{10}} = \sqrt{10}$.

45 [Challenging] • Target: Position, velocity, and acceleration vectors • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Stokes' theorem for $\mathbf{F} = (y,\ -x, 0)$ over unit disk $z = 0$: $\oint \mathbf{F} \cdot d\mathbf{r} =$

a. 0

b. 2π

c. -2π ✓

d. π

$abla \times \mathbf{F} = (0, 0, -2) \Rightarrow \iint (0, 0, -2) \cdot (0, 0, 1) dA = -2\pi$.

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